

AI Comic Story Creator Using Gemini Models

Project Documentation

Project Overview

AI Comic Story Creator is a web-based application that helps users create comic stories using Google Gemini AI models. Users can enter a theme, genre, characters, and a short story idea. The application then generates a comic storyline, scene-wise dialogue, narration, and image prompts that can be used to create comic panels.

The goal of this project is to make comic creation easier, faster, and more accessible for students, writers, and creative users without requiring advanced drawing or writing skills.

1. Brainstorming & Ideation Phase

The initial idea was to create an AI-powered platform that transforms a simple user idea into a complete comic story. Many people have creative ideas but find it difficult to write stories, develop characters, or design comic scenes.

The proposed system uses Gemini models to generate:

- Story titles and themes
- Character descriptions
- Story plots
- Scene-by-scene comic scripts
- Dialogues and narration
- Image prompts for comic panels

The main idea is to combine artificial intelligence and creativity in a simple application that allows users to generate personalized comic stories.

2. Requirement Analysis Phase

Functional Requirements

- Users can enter a comic topic or story idea.
- Users can select a genre such as action, fantasy, comedy, horror, romance, or science fiction.
- Users can define characters and their personalities.
- The system generates a story title and storyline using Gemini AI.
- The system divides the story into multiple comic scenes.
- The system generates dialogue and narration for every scene.
- The system creates image prompts for comic-panel generation.
- Users can edit generated content before saving or exporting it.
- Users can download or share the generated comic story.

Non-Functional Requirements

- The application should provide results quickly.
- The user interface should be simple and easy to use.
- User inputs and generated stories should be handled securely.
- The system should support multiple genres and creative writing styles.

- The application should be scalable for future features such as image generation and multilingual support.

Hardware and Software Requirements

- Computer with internet connection
- Web browser
- Python / JavaScript development environment
- Google Gemini API key
- Database for storing user data and generated stories

3. Project Design Phase

The project follows a modular architecture.

Main Modules

- 1. **User Input Module** – Collects story topic, genre, characters, setting, and number of scenes.
- 2. **Gemini AI Integration Module** – Sends prompts to the Gemini API and receives generated story content.
- 3. **Story Generation Module** – Creates the title, plot, character details, dialogues, narration, and scene sequence.
- 4. **Comic Panel Module** – Converts each story scene into structured comic-panel content and image prompts.
- 5. **Editor Module** – Allows users to modify generated dialogues, scenes, and character details.
- 6. **Export Module** – Enables users to save or download the comic story in text or PDF format.

Technology Stack

- **Frontend:** HTML, CSS, JavaScript / React
- **Backend:** Python Flask / FastAPI or Node.js
- **AI Model:** Google Gemini API
- **Database:** MongoDB / Firebase / MySQL
- **Hosting:** GitHub Pages, Render, Vercel, or Firebase

4. Project Planning Phase

The development plan is divided into the following stages:

Week	Activity
Week 1	Finalize project idea and collect requirements
Week 2	Design user interface and system architecture
Week 3	Set up Gemini API integration
Week 4	Develop story and dialogue generation modules
Week 5	Develop comic scene and image-prompt generation
Week 6	Test application features and fix errors
Week 7	Prepare documentation and project report
Week 8	Demonstrate the completed project

Team Responsibilities

- Requirement collection and planning
- Frontend development

- Backend and API integration
- Testing and debugging
- Documentation and presentation

5. Project Development Phase

During development, the application was built with a user-friendly interface where users can provide their comic idea and preferences. The Gemini model is integrated through an API to generate creative content based on the user prompt. The system creates a structured response containing:

- Comic title
- Genre
- Character profiles
- Story summary
- Scene descriptions
- Dialogues
- Narration
- Image prompts

The generated output is displayed as comic scenes, allowing users to review and edit the story before downloading it.

Sample Input

Genre: Science Fiction
Theme: A young inventor discovers a robot on Mars.
Characters: Arjun, Nova the Robot
Number of Scenes: 5

Sample Output

Title: The Lost Robot of Mars

Scene 1:

Arjun lands on Mars and discovers an abandoned robot near an old research station.

Dialogue:

Arjun: "Who left you here?"

Nova: "I have been waiting for someone to find me."

6. Project Testing Phase

The application was tested to ensure that all modules work correctly.

Test Case	Expected Result
User enters a valid story idea	System generates a comic story
User selects a genre	Generated story follows the selected genre
User enters character details	Characters appear in the story
Gemini API request is successful	Story content is returned and displayed
Invalid or empty input is submitted	System shows validation message
User edits generated content	Updated content is saved correctly
Export option is selected	Comic story is downloaded successfully

Testing Outcome

The application successfully generates creative comic stories based on user input. Input validation and error handling are included to improve reliability. API errors, invalid prompts, and empty fields are handled with appropriate user messages.

7. Project Documentation Phase

The project documentation includes:

- Project title and abstract
- Problem statement
- Objectives
- Requirement analysis
- System design
- Technology stack
- Module descriptions
- API integration details
- Testing results
- Screenshots
- Future enhancements
- Conclusion

Future Enhancements

- Generate comic images directly using an image-generation model.
- Add support for Tamil and other regional languages.
- Allow users to choose comic art styles such as manga, cartoon, realistic, or superhero.
- Add voice narration for comic dialogues.
- Enable cloud storage and sharing of comic projects.
- Add collaborative story creation for multiple users.

8. Project Demonstration Phase

During the project demonstration, the following workflow will be shown:

1. Open the AI Comic Story Creator application.
2. Enter a comic idea, genre, and character details.
3. Submit the prompt to Gemini AI.
4. Display the generated title, plot, scenes, and dialogues.
5. Show image prompts generated for each comic scene.
6. Edit a scene or dialogue.
7. Download or export the completed comic story.

Conclusion

AI Comic Story Creator demonstrates how generative AI can support creative storytelling. By using Gemini models, the system converts simple ideas into structured comic stories with characters, dialogues, and scene descriptions. The project reduces the effort required for story creation and encourages users to explore their imagination through

AI-assisted comic development.